

COMP307: Project Specification

Project Objectives

The aim of the project is for learners to apply the graphics and modelling principles covered in this module. Students will create a 3D application, typically a real-time game, using a graphics library or engine that allows them to demonstrate these core concepts. The focus should remain on applying transformation, modelling, and rendering techniques.

Game Specification

The project involves the design and implementation of an advanced game application made using a gaming engine: `Unity` **is recommended**. Groups may alternatively use `OpenGL` exclusively if they have the technical proficiency to do so.

The game must contain the following:

- One or more playable characters that are able to move around and interact with the environment.
- Non-playable characters (NPCs) that have defined behaviour in the world.
- Win and loss conditions.

Additional Information

- The use of prefabs is acceptable, but the game must demonstrate that the group is able to apply the concepts learnt in the module. This must be shown by including some characters or assets that are **modelled from first principles**.
- You are also required to submit the source code of your game to ensure reproducibility. The best approach would be to create a `Github` repository.
- In addition, each group is required to write at least one **custom shader using `Unity ShaderLab (HLSL)` or `GLSL`**. This shader must demonstrate one or more graphics techniques covered in the module, such as lighting models, surface properties, vertex animation, etc.
- The above specifications must be adhered to; however, students are encouraged to go above and beyond them and will be awarded marks should they do so.
- This is a group project that will be completed in groups of 4. Working in teams is an essential soft skill that you need to develop. One member from each group must send me an email containing the names and student numbers of each member in your group by **Friday, 08 August 2025**. This project cannot be completed individually!
- You are allowed (and encouraged) to create groups that contain members from both WVL and PMB campuses.

Deliverable Structure

The project will be undertaken using an iterative development approach. This means that a number of deliverables will be assessed during the progression of the project, culminating in the assessment of the final deliverable at the end of the course. The deliverable breakdown for the project is:

1. Game Concept and Overview
2. First Prototype
3. Final Deliverable

Deliverable 1: Game Concept and Overview
Due: Friday, 22 August 2025

This document provides an overview of the game. Careful planning is an essential step that must be undertaken before the development of the game (or any project) begins. The success of the project depends a great deal on the effort that is put in at this early stage. As a guideline, the document may contain the following details/headings: (Note that some headings may not even be applicable to your game idea)

1. Game Name
2. Introduction
    - What is the purpose of the game? What is there to do in the game? What is the game actually about?
    - Describe the camera angle(s) and aesthetic of the game. Does the camera look down on the main character? Is the game from a first-person perspective?
3. Gameplay overview
    - Describe the flow of gameplay. How does the player interact with the world? What are the purpose and behaviour of the other characters that he/she can interact with?
    - Give an overview of how the game plays out.
4. World Layout
    - Describe the layout and shape of your world. How do the player and other characters move along it?
    - Which assets are created from First principles?
5. Character Descriptions
    - Describe any characters in the game and their capabilities. This includes the player and non-player characters. How do these characters move around the world? For example: are they able to shoot? Can they fly around or do they only move along the ground?
    - Which characters are created from First principles?
6. Controls
    - How do the user(s) control their character(s) and interact with the world?
7. Win/Loss Conditions
    - Describe all win/loss conditions. What conditions cause the game to end?

Deliverable 2: First Prototype
Due: Friday, 19 September 2025

Each group must prepare and present a 10-minute presentation that must include:

1. A demonstration of a WORK IN PROGRESS, RUNNING version of your game, touching on the different aspects of the Marking Memo:
   (a) Movement, world interaction and mechanics
   (b) NPCs and/or basic AI
   (c) Use of Shaders
   (d) Uniqueness and/or (proposed) Extensions
   (e) Win/loss conditions
2. Remaining work with a proposed timeline on how the group plans to complete the project
3. Any challenges that the group envisages

Deliverable 3: Final Deliverable
Due: Wednesday, 15 October 2025

Each group must submit the following:

1. A presentation (10-15 minutes including questions) that clearly details the game's design, implementation, and graphical techniques used: you will need to touch on the different aspects of the Marking Memo. The presentation must include:
   - A brief gameplay demo
   - Key design decisions
   - Reference to original models
   - An explanation of the custom shader(s) developed and how they demonstrate concepts from the module.
2. Source code for the final version of the game, fully functional and packaged for demonstration. This must include a README detailing how to run the code and be submitted in a zipped folder or via version control (e.g., Git).
3. The slides used for your presentation.
4. An anonymous self and buddy evaluation touching on each group member's contribution